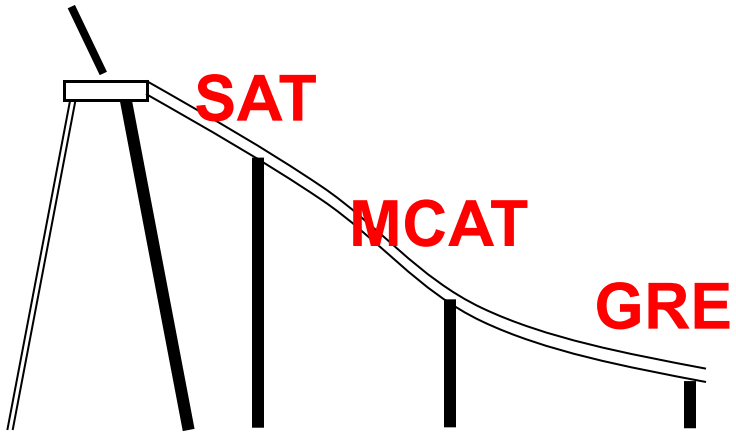
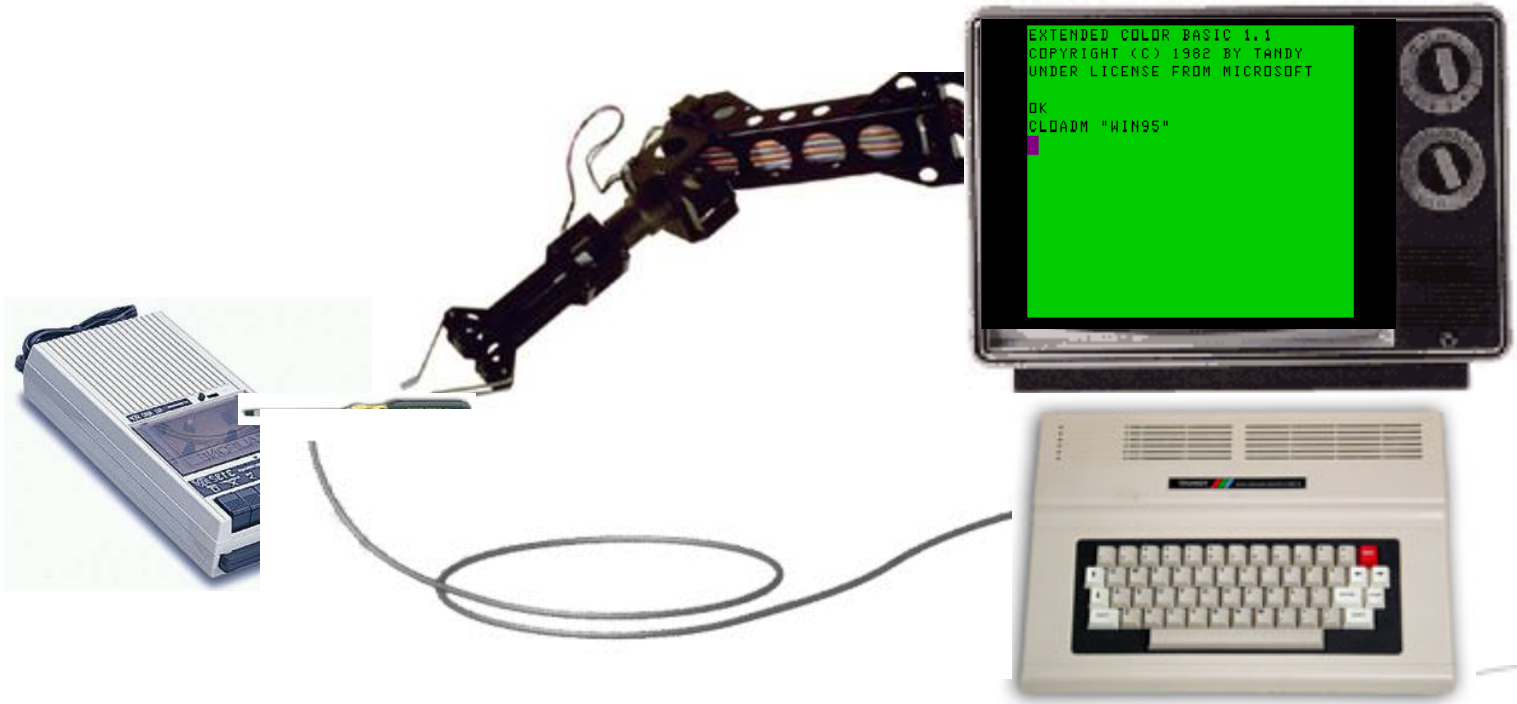
Test Slide



Quantum Computers That Fix Themselves



Dave Bacon
Caltech
Department of Physics
Institute for Quantum Information

?





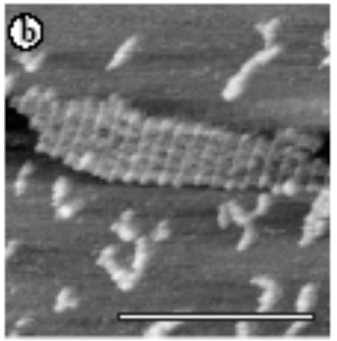
Red or Green?

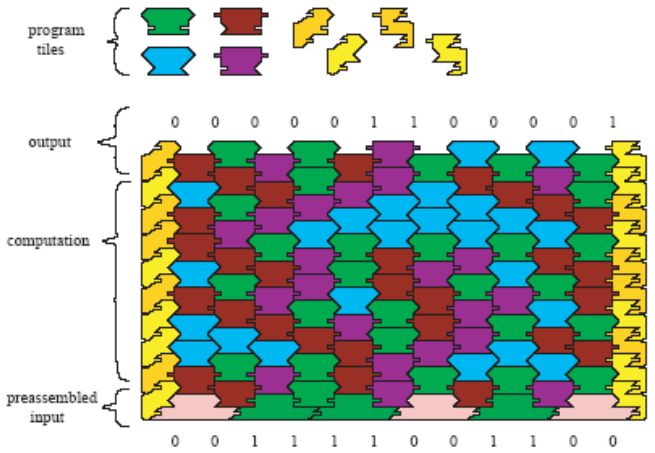
Unanswerable

Is it a Computer?



Ship Rock



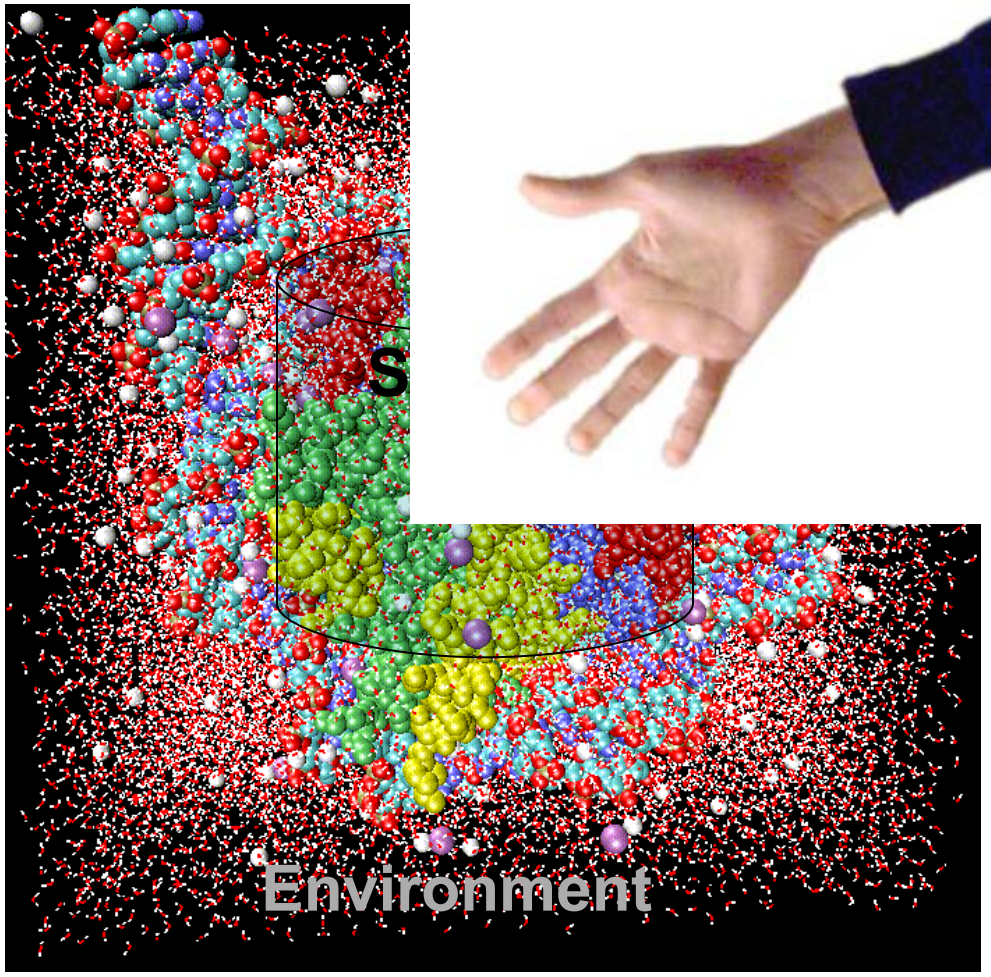


DNA Universal Computing

(pics from Eric Winfree's lab)

Is it a ROBUST Computer?

The Computer Phase





Control

- Controllable yet Decoupled
- Switching, Digital
- Regular yet Complex

Reducible yet Irreducible

Computerology

Computation Is Physical



Information Carriers Are Physical

Rolf Landauer, IBM

Physics is the beginning (but not nearly the end)
of **computerology**

Quantum Computing

"Perhaps…we need a mathematical theory of quantum automata…" – Yu Manin (Radio Moscow, 1980)

"How can we simulate quantum mechanics?...Can we do it with a new kind of computer…a quantum computer? It's not a Turing machine but a machine of a different kind" – Richard Feynman (1981)

0
1 bits



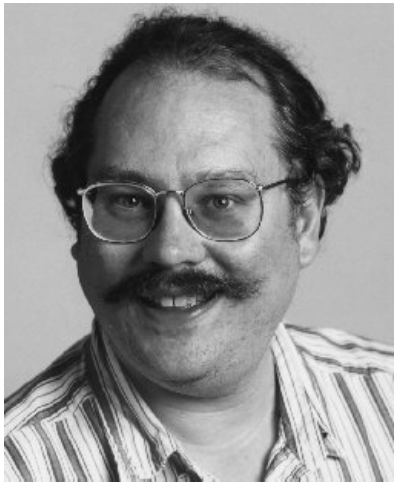
$|0\rangle$
$|1\rangle$ qubits

32% **0** 68% **1**

$\frac{1}{\sqrt{2}}|0\rangle - \frac{i}{\sqrt{2}}|1\rangle$

A computer whose state is quantum information and whose evolution obeys the laws of quantum theory

The Hopes



Peter Shor (1994) – An efficient algorithm for factoring numbers using a quantum computer.

Breaks RSA Cryptography



“Only time will tell if and when the problems of building a quantum computer can be overcome….As information becomes the world’s most valuable commodity, the economic, political and military fate of nations will depend on the strength of ciphers. Consequently, the development of a fully operational quantum computer would imperil our personal privacy, destroy electronic commerce and demolish the concept of national security. A quantum computer would jeopardise the stability of the world. Whichever country gets there first will have the ability to monitor the communications of its citizens, read the minds of its commercial rivals and eavesdrop on the plans of its enemies. Although it is still in its infancy, quantum computing presents a potential threat to the individual, to international business and to global security. “
-Simon Singh

More Hopes

…

Hallgren (2002) – An efficient quantum algorithm for Pell's equation and principle ideal problem

Hallgren, Ip, van Dam (2003) – Quantum algorithms for some hidden shift problems.

Moore, Rockmore, Russell, Schulman (2003) – Quantum algorithm for the hidden subgroup problem in affine groups.

…



Ran Raz (1999)



x

F(x,y)

y



ALICE

BOB

how much communication?

Exponentially less quantum communication than classical communication!

The Big Themes

WHAT can we do with a quantum computer?

WHY is a quantum computer algorithmically powerful?

How can quantum information science influence the best of science?

HOW can we build a quantum computer?

BIG IDEAS

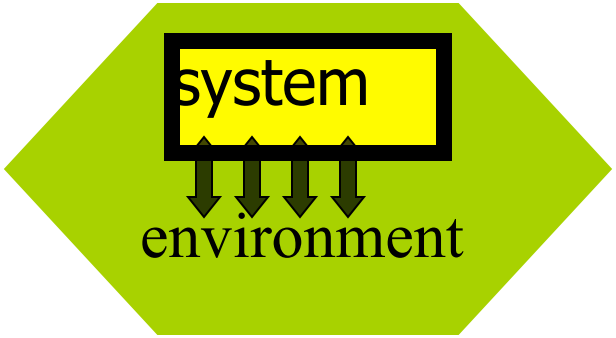
One Method?



"And this [image] th of the ark shall be three hundred cubits, the breadth of it fifty cubits, and the height of it thirty cubits." – Genesis 6:15

What is the Problem?

Real quantum systems are open quantum systems!



Quantum systems readily couple to an environment…

System decoheres:

$|0\rangle$ qubits

$|1\rangle$

$\frac{1}{\sqrt{2}}|0\rangle - \frac{i}{\sqrt{2}}|1\rangle$

Quantum

0 bits

1

50% **0** 50% **1**

Classical

The Decoherence Problem (1996)

What Is The Problem?

Decoherence is a lot like classical noise, BUT:

Strong coupling to control devices needed to enact computations



Strong coupling to environment causes decoherence

Yingyang of quantum computing

Quantum Computing is Bunk

Ways Quantum Computers Fail to Quantum Compute

Decoherence

$$(\alpha|0\rangle + \beta|1\rangle) \otimes |\text{environment}\rangle \rightarrow \alpha|0\rangle \otimes |\text{environment}_A\rangle + \beta|1\rangle \otimes |\text{environment}_B\rangle$$

Lack of Unitary Control

attempting to apply unitary evolution **U** instead results in **V**

or (worse) results in non-unitary evolution

$$\boldsymbol{\rho} \rightarrow \sum_k \mathbf{A}_k \boldsymbol{\rho} \mathbf{A}_k^\dagger$$

Measurements are faulty

measurement result is noisy, incorrect result obtained

Quantum Computing Disappearing Act

qubits disappear (leakage of computing states)

A Model Solution



Alan Turing

Decoherence is a lot like classical noise

Turing was crazy!
Reliable components for a computer?

Theory History

Shannon (1948): error correcting codes

von Neumann (1956): how to perform fault-tolerant classical computation

Quantum Computing (1994): Can we construct a theory of fault-tolerant quantum computation?

1994 Reasons to be a Pessimist

No cloning:

No cloning of quantum information:

$$\nexists \mathcal{T} || \psi \rangle \rightarrow |\psi\rangle \otimes |\psi\rangle, \forall |\psi\rangle$$

Quantum repetition code impossible:

$$|\psi\rangle \nrightarrow |\psi\rangle \otimes |\psi\rangle \otimes |\psi\rangle$$

Continuous set of errors:

Classical errors on digital information are discrete.

Quantum errors on classical information are continuous.

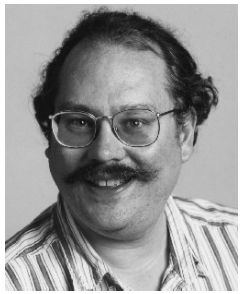
Continuous set of operations:

Continuous set of unitary operations: infinite precision needed to implement a gate?

Measurement destroys coherence:

How can one decode without destroying the information?

The Quantum Solution (1995-96)



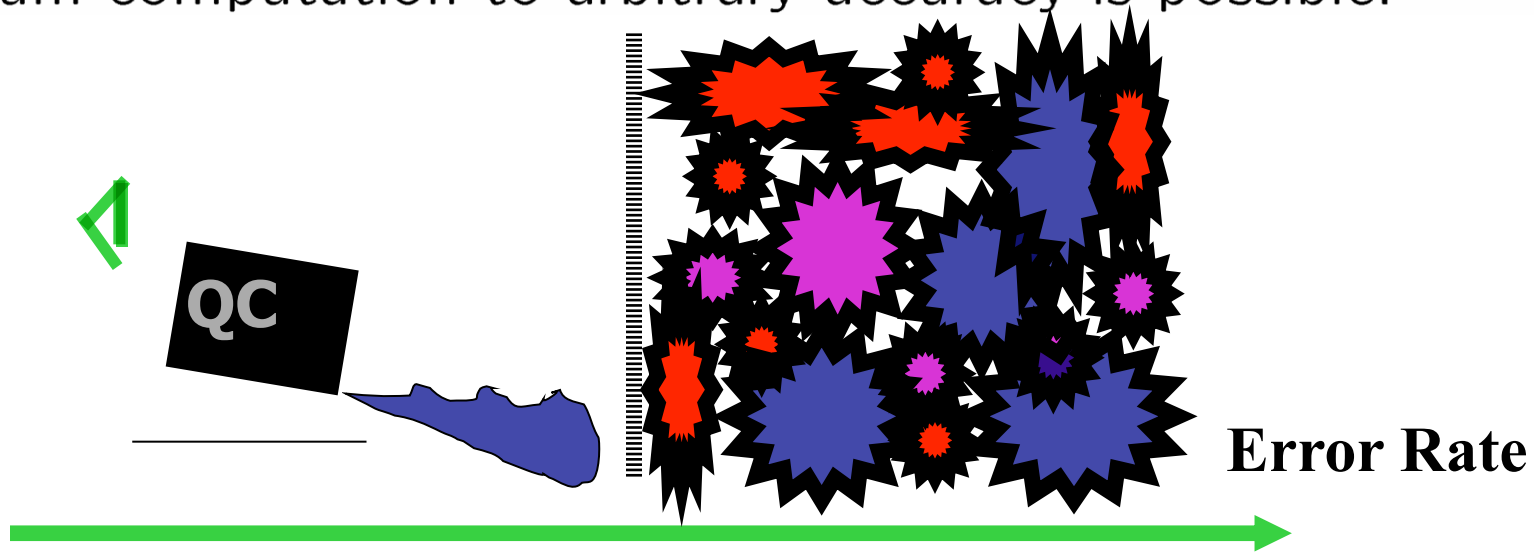


Peter Shor Andrew Steane

Fault Tolerant Quantum Computing is Possible!

Threshold Theorem:

If the error rate (typical number of operations per decoherence or failure event) for single qubit errors is smaller than ϵ, then quantum computation to arbitrary accuracy is possible.



1941 – First programmable electronic calculator.
mechanical relays.
1943 – ENIAC.
18000 vacuum tubes: “nature abhors the vacuum tube.”
1947 – Bell labs develops the transistor.
1952 – G. W. Dummer proposes manufacturing electronic equipment in one block with no connecting wires.
1959 – Texas Instruments and Fairchild Semiconductor invent the integrated circuit.
1964 – first IBM 360 series (the concept of an “architecture”)
1964 – Integrated circuit which cost $1000 in 1959 now costs $10. Moore describes his law.
1972 – Intel 8008 and 8 bit microprocessor.
1975 – First personal computer the Altair.
1982 – IBM PC introduced. Personal computer revolution begins.

Have We Learned Anything?

Rant mode ON

The Physics Guarantee

What is the phase of matter corresponding to the computer?

PRACTICAL QUESTION (as opposed to philosophical)

There are distinct PHYSICAL and DYNAMICAL reasons why robust classical computation is possible.

not all physical systems are equally good for computation:
there exists systems whose PHYSICS guarantees their ability to enact robust classical computation.

Are there (or can we engineer) physical systems whose PHYSICS guarantees robust quantum computation?

This talk: (1) this point of view for classical computers
(2) attempts to port these ideas to a quantum computer

Two Paths ("it's made all the difference")

Hard Drive



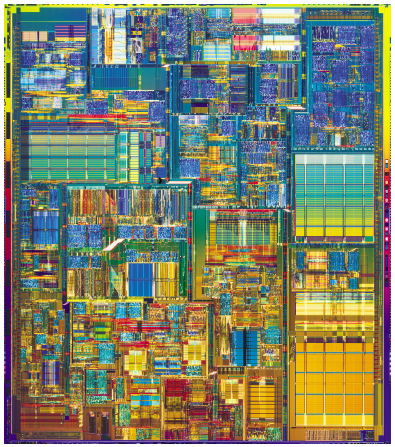
Coding: majority vote of magnetism

Error correction: local energy minimization

Fault-tolerant: macro applied field.

More than storage?

Transistor



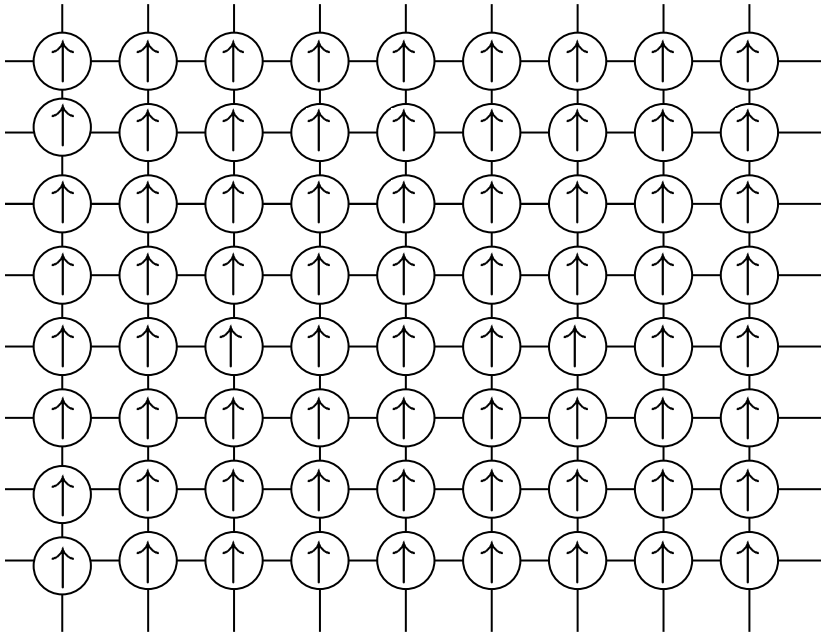
Coding: majority vote of current

Error correction: amplification

Fault-tolerant: conducting / insulating phase transition

A Dynamic Ising Model





Ising model

spins on a lattice $s_i \in \{+1, -1\}$

$$E = -J \sum_{\langle ij \rangle} s_i s_j$$

Energetically favorable to align

Order parameter:

$$M = \sum_i s_i$$

(the information)

Dynamic Ising model

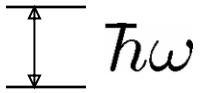
interaction with environment causes spins to flip



Reminder



two-level system radiatively coupled to a thermal reservoir:



$$\frac{\partial \boldsymbol{\rho}}{\partial t} = -\bar{n}_{th}\frac{\gamma}{2}\left[\boldsymbol{\sigma}_-\boldsymbol{\sigma}_+\boldsymbol{\rho} + \boldsymbol{\rho}\boldsymbol{\sigma}_-\boldsymbol{\sigma}_+ - 2\boldsymbol{\sigma}_+\boldsymbol{\rho}\boldsymbol{\sigma}_-\right]$$

"heating"

$$-(\bar{n}_{th}+1)\frac{\gamma}{2}\left[\boldsymbol{\sigma}_+\boldsymbol{\sigma}_-\boldsymbol{\rho} + \boldsymbol{\rho}\boldsymbol{\sigma}_+\boldsymbol{\sigma}_- - 2\boldsymbol{\sigma}_-\boldsymbol{\rho}\boldsymbol{\sigma}_+\right]$$

"cooling"

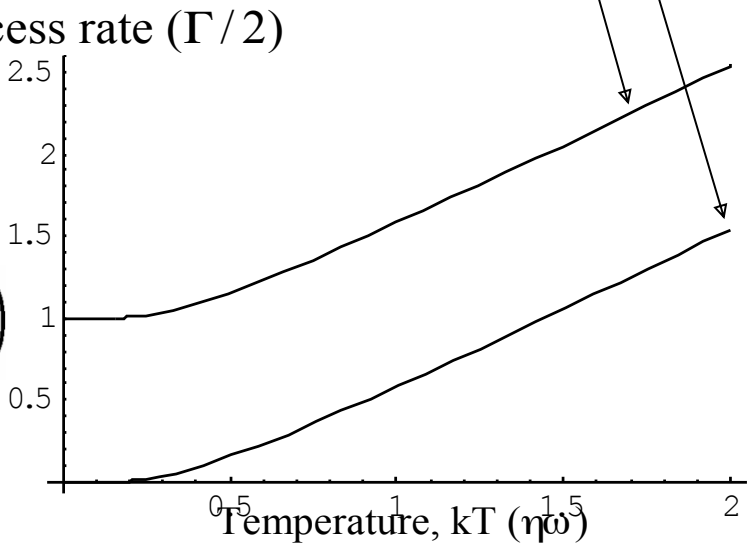
$$-\phi\left[\boldsymbol{\rho} - \boldsymbol{\sigma}_z\boldsymbol{\rho}\boldsymbol{\sigma}_z\right]$$

"non-dissipative"

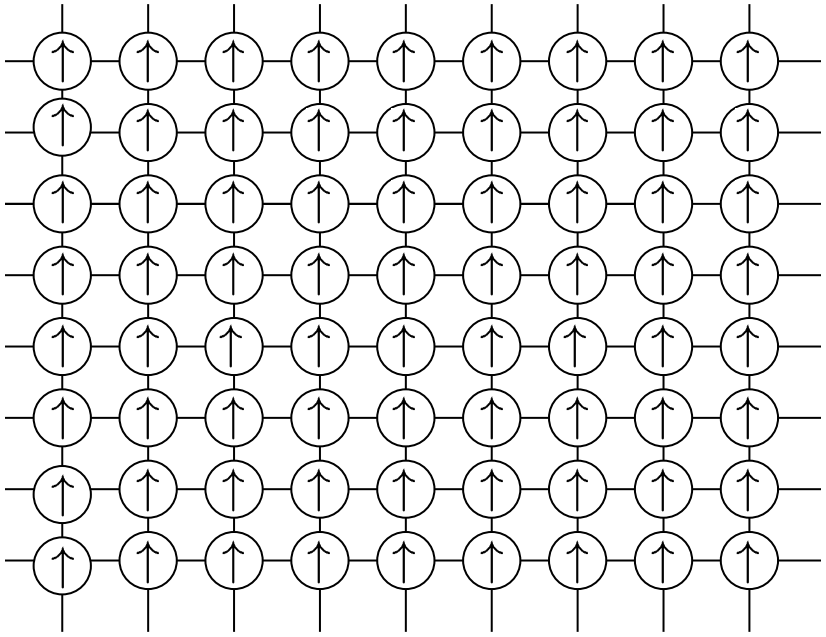
$$\bar{n}_{th} = \frac{1}{\exp\left(\frac{\hbar\omega}{kT}\right)+1}$$

$$\frac{\text{heating}}{\text{cooling}} = \frac{\bar{n}_{th}}{\bar{n}_{th}+1} = \exp\left(-\frac{\hbar\omega}{kT}\right)$$

$$= \exp\left(-\beta\Delta E\right)$$



A Dynamic Ising Model



Ising model

spins on a lattice $s_i \in \{+1, -1\}$

$$E = -J \sum_{\langle ij \rangle} s_i s_j + h \sum_i s_i$$

Energetically favorable to align

Order parameter:

$$M = \sum_i s_i$$

Dynamic Ising model (Metropolis update)

interaction with environment causes spins to flip

for every spin

if flipping spin would decrease energy, flip spin

if flipping spin would increase energy, flip spin with probability $\exp(-\beta \Delta E)$

Metropolis Monte Carlo

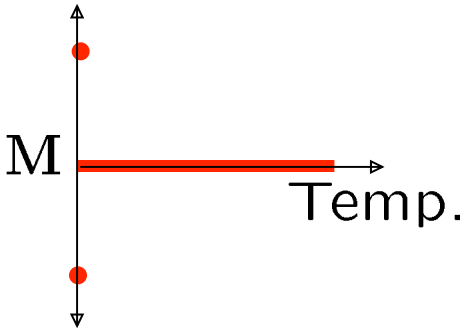
Storage and Manipulation

Compare 1D and 2D Ising models at different temperatures for
a) storage of information
b) manipulation of information

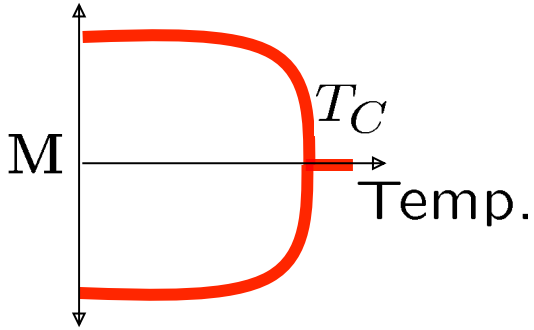
1st attempt:

Storage in thermal state (infinite time - ensemble is thermal)

1D Ising



2D Ising



Criticism: (1) thermodynamic limit taken (N infinite)
(2) what if relaxation to equilibrium takes a long time?

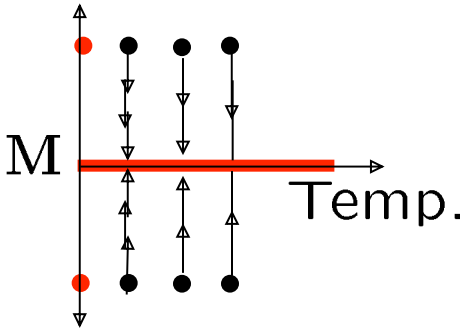
Storage



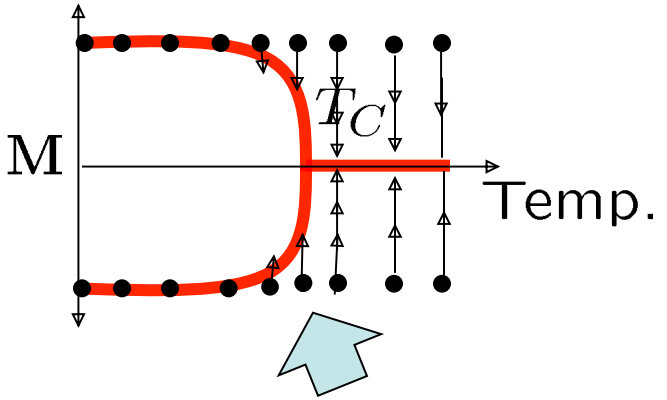
2nd attempt:

Relaxation to thermal

1D Ising



2D Ising

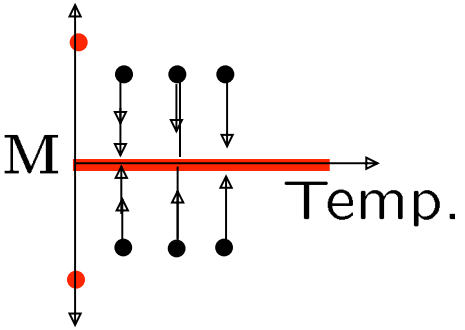


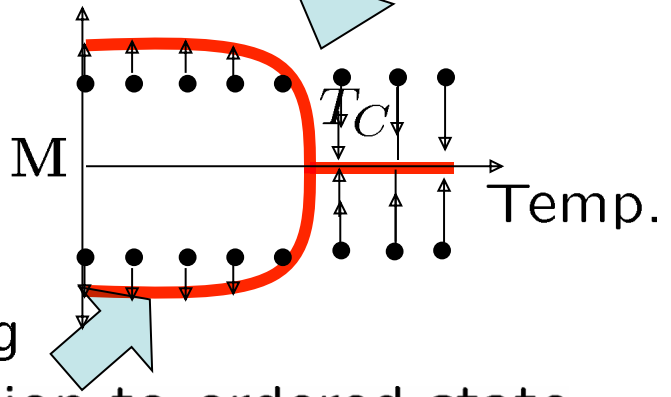
critical slowdown

$t_{th} \propto |T - T_c|^{z-\nu}$

decay rate $\propto e^{-\beta\Delta}$

Imperfect preparation





self correcting

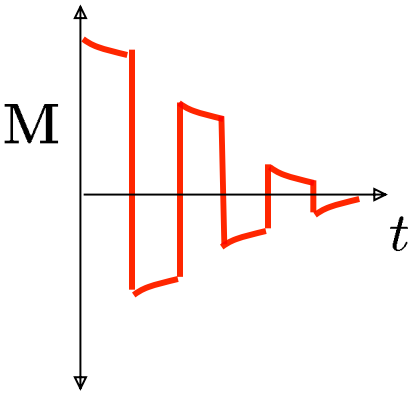
fast relaxation to ordered state

Manipulation

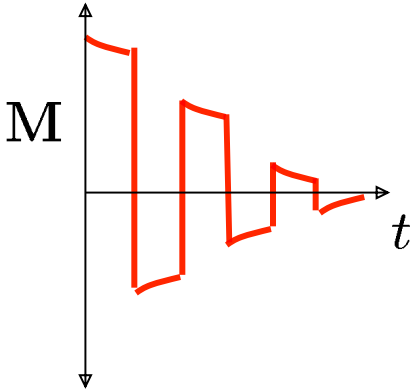


Flipping spins is imperfect

1D Ising

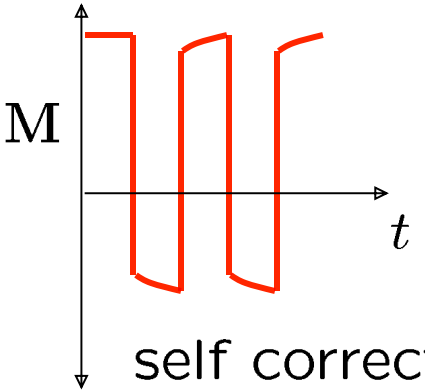


2D Ising



$T > T_C$

1D Ising
- temperature less than gap
- manipulation error must be small

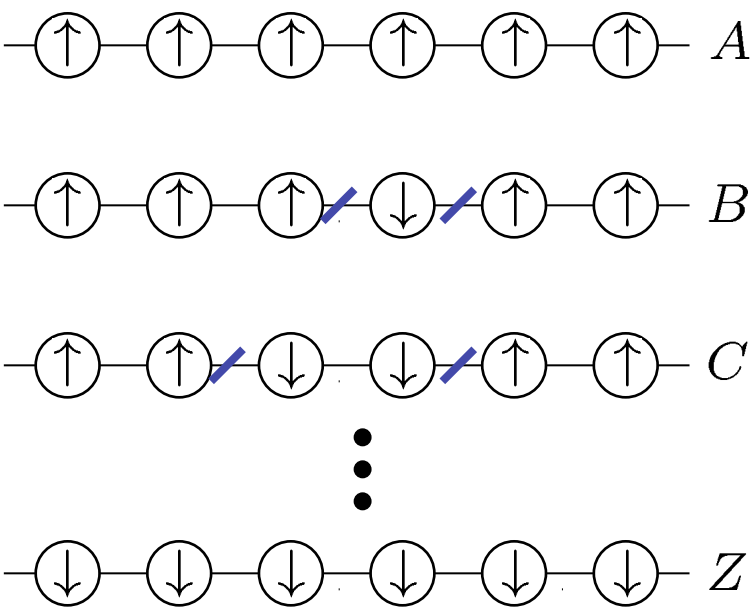


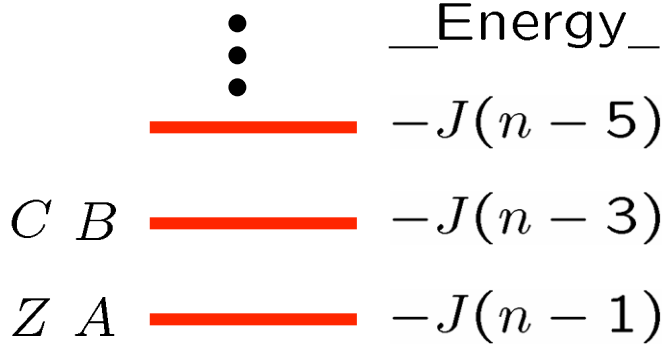
$T < T_C$

2D Ising
- temperature less than critical temp
- self correcting

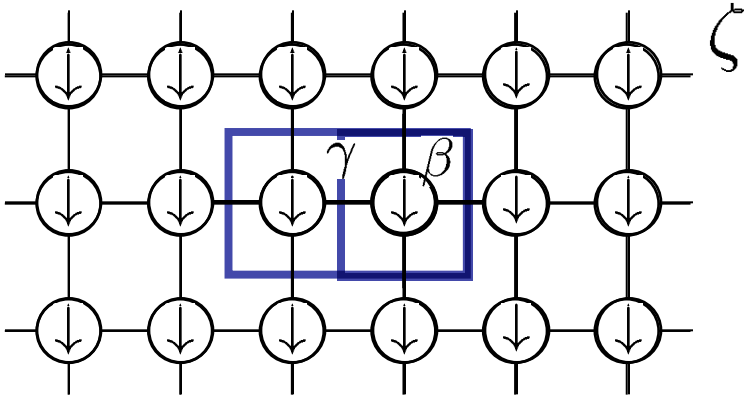
self correcting

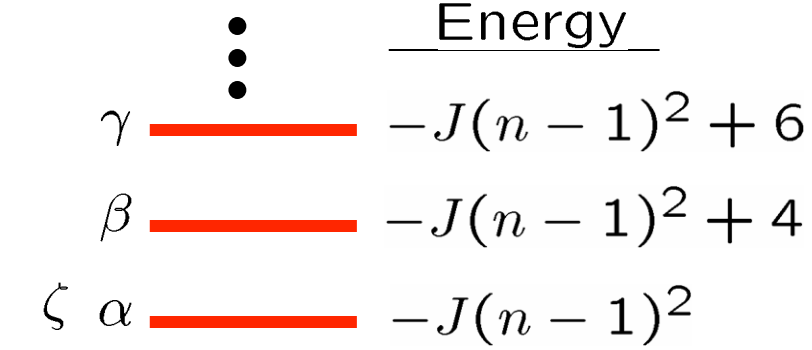
Self Fixing





Excitations free pointlike.
Energy cost constant.
disordering entropy $\gg$ energy





Excitations stringlike.
Energy cost grows like perimeter.
energy $\gg$ disordering entropy

Two Paths

Hard Drive



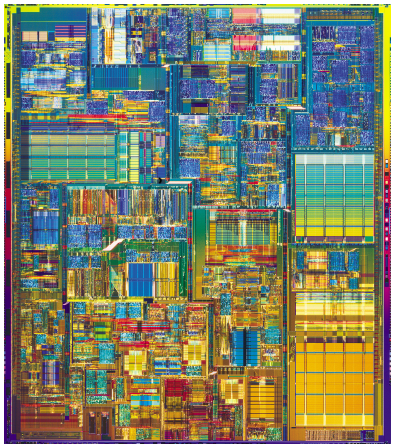
Coding: majority vote of magnetism

Error correction: local energy minimization

Fault-tolerant: macro applied field.

More than storage?

Transistor

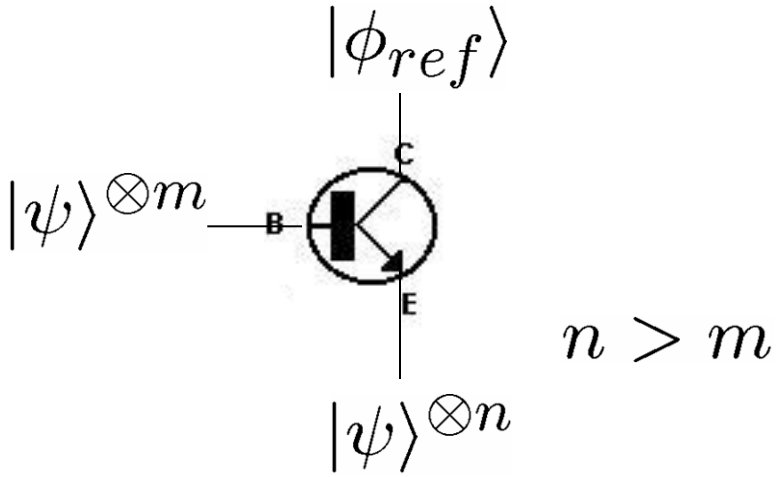


Coding: majority vote of current

Error correction: amplification

Fault-tolerant: conducting / insulating phase transition

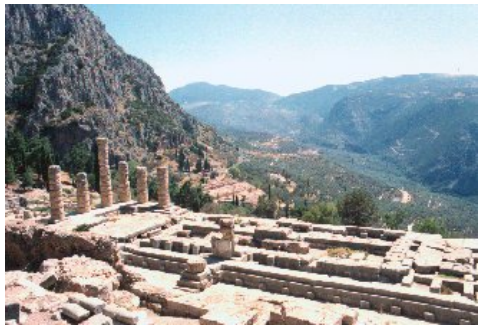
No Quantum Transistor





Quantum Cloning Machine

"A single quantum cannot be cloned," Wootters and Zurek, Nature, 1982

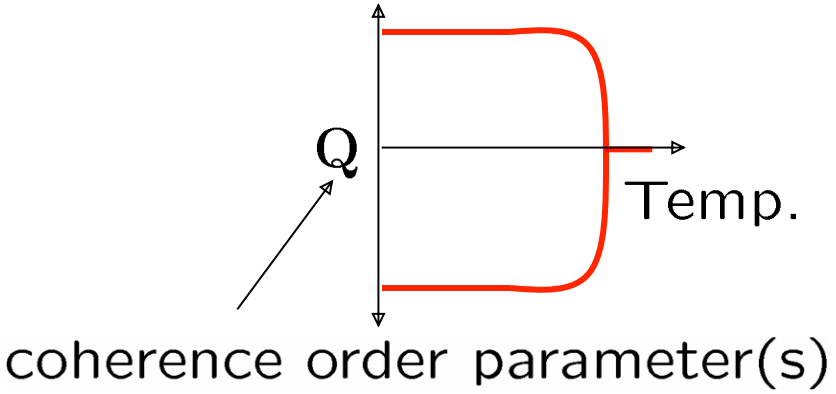


The Quantum Hard Drive?

(Kitaev)



Do there exist (or can we engineer) quantum systems whose physics guarantees fault-tolerant quantum computation?



1. Coherence preserving.

$Q_1 \Rightarrow -Q_1$

macro local operations

2. Accessible Fault-Tolerant Operations

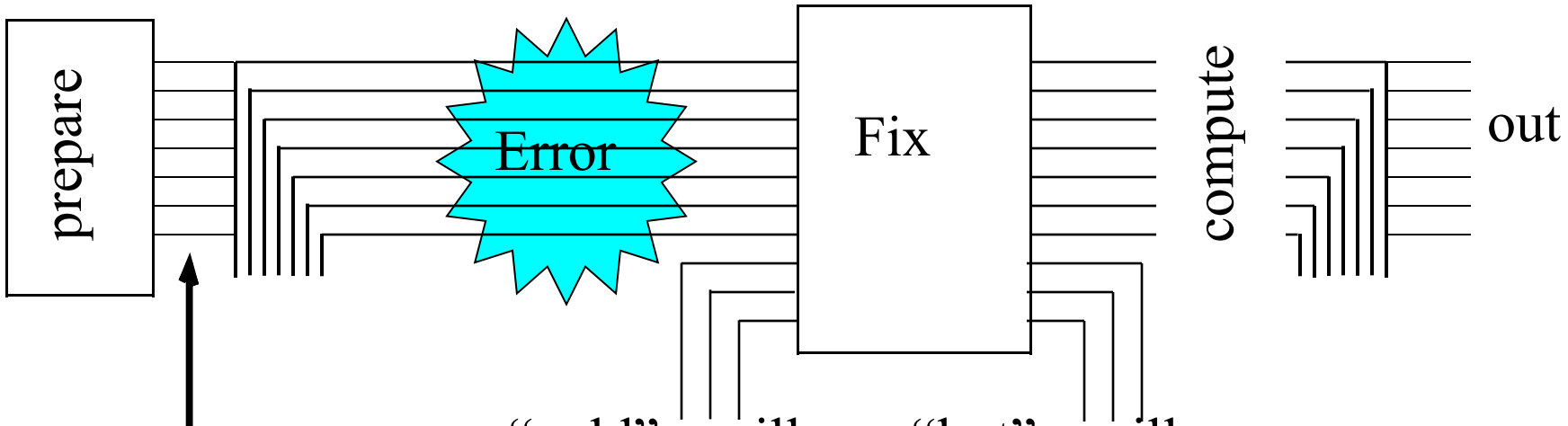
"self-correcting"

$$\left\langle G_1, \ldots, G_{n^k} \right\rangle \cong U(c^n)$$

3. Universality

Old School

"Information is fungible!"



Quantum error correcting code

$$\mathcal{H} \subset \mathcal{V}^{\otimes n}$$

$$|i_L\rangle = \sum_\alpha c_{\alpha,i}|\alpha\rangle, \alpha \in \mathcal{V}^{\otimes n}$$

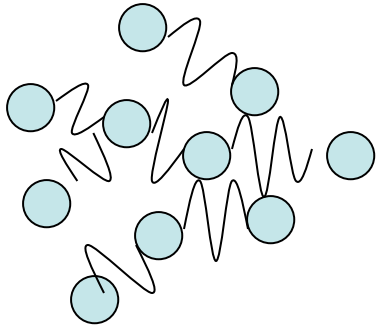
Competition between errors a cooling

Error correcting criteria:

$$\langle i_L|\mathbf{E}_b^\dagger \mathbf{E}_a|j_L\rangle = c_{ab}\delta_{i,j}$$

If criteria holds, then fixing procedure possible

Into The Fray



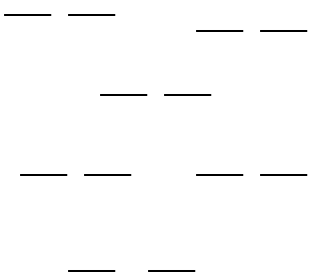
Thermodynamic limit

$$N \to \infty$$

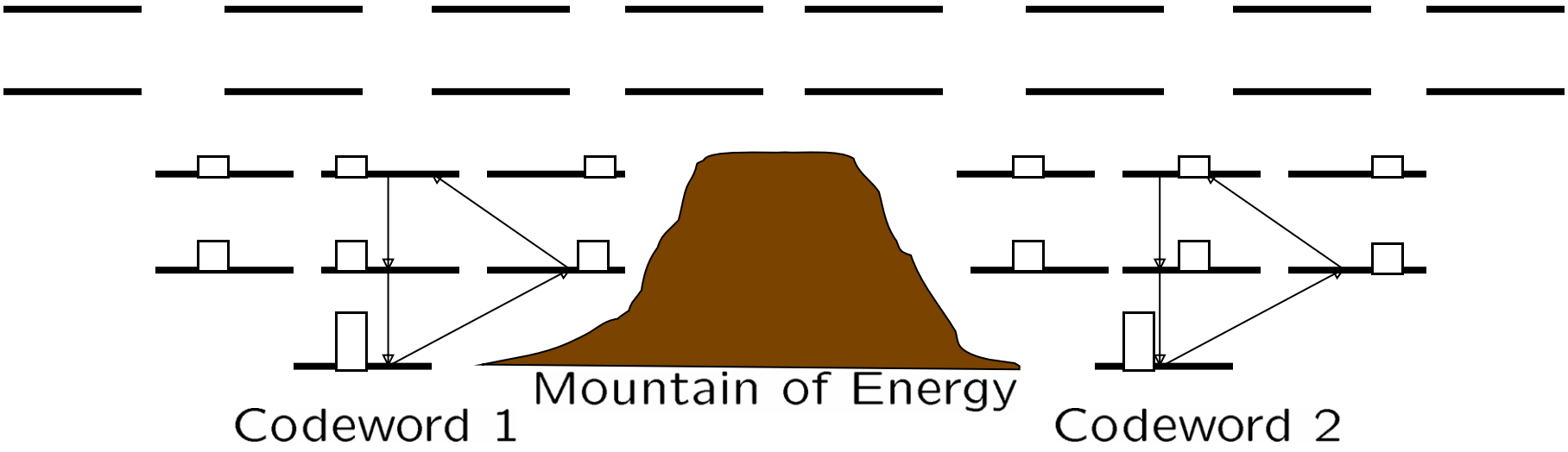
guarantees robustness

Strongly Interacting Many Body Quantum System

Hamiltonian $\mathbf{H}$

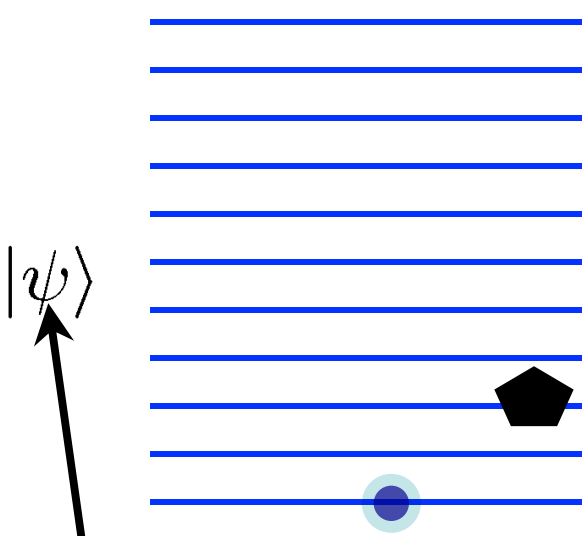


energy eigenstates

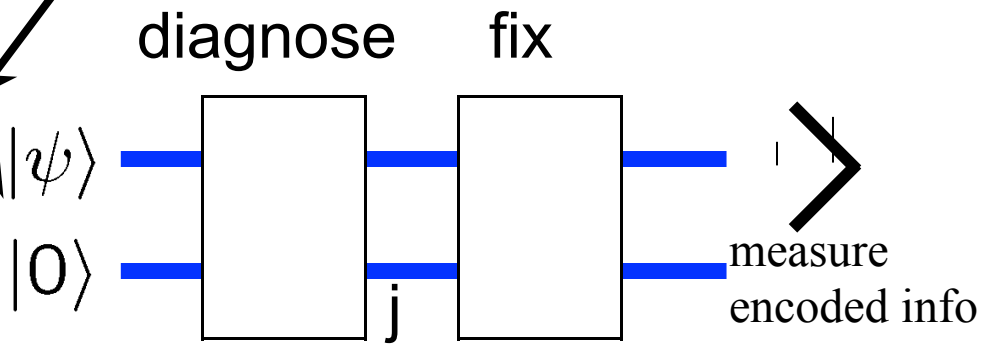


Barnes and Warren, PRL 85, 856 (2000) [but is only “classical”]

Quantum Error Correcting Order Parameters



The quantum information is still here. How do we see it?



Encoded quantum information

$$|\psi\rangle = \alpha|0_L\rangle + \beta|1_L\rangle$$

$$\mathbf{Q}_s = \sum_j \mathbf{L}_s \mathbf{F}_j \mathbf{D}_j$$

$\mathbf{L}_s$ logical encoded operation

$\mathbf{D}_j$ projector onto jth errored subspace

$\mathbf{F}_j$ fixing operator for jth errored subspace

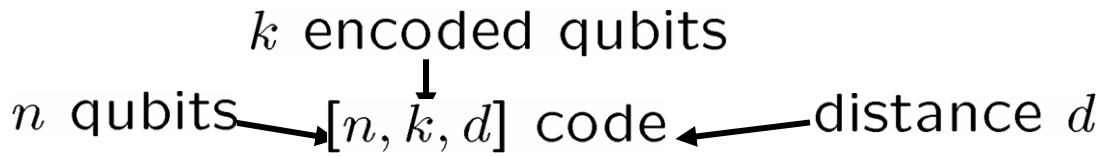
Naturally Fault-Tolerant Quantum Computers Will Have New Phases and New Order Parameters!

A Simple Start

"To be an Error and to be Cast out is Part of [Nature's] Design."

-*William Blake*

$$\mathbf{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \mathbf{Y} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \mathbf{Z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$



[4,2,2] quantum error detecting code:

$$\mathbf{S}_1 = \mathbf{XXXX} \quad \mathbf{S}_2 = \mathbf{ZZZZ} \qquad \mathbf{S}_i|\psi\rangle = |\psi\rangle$$

Encoded operations on encoded ($\mathbf{S}_i|\psi\rangle = |\psi\rangle$) two qubits

$$\mathbf{X}_1 = \mathbf{XXII},\ \mathbf{Z}_1 = \mathbf{IZZI}$$

$$\mathbf{X}_2 = \mathbf{IXXI},\ \mathbf{Z}_2 = \mathbf{ZZII}$$

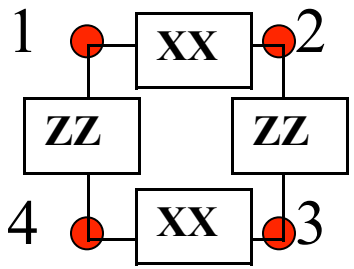
Complete basis: $|S_1, S_2, Z_1, Z_2\rangle,\ S_i, Z_i \in \{\pm 1\}$

$\mathbf{PS}_i = -\mathbf{S}_i\mathbf{P}$ for some i

$\langle i_L|\mathbf{P}|j_L\rangle = 0$, $\mathbf{P}$ single qubit operator

Supercoherence

4 qubit system



$$S_1 = XXXX$$
$$S_2 = ZZZZ$$
$$X_1 = XXII,\ Z_1 = IZZI$$
$$X_2 = IXXI,\ Z_2 = ZZII$$

System Hamiltonian: $H = XXII + IIXX + ZIIZ + IZZI$

Note symmetries: $[H, S_i] = [H, X_2] = [H, Z_2] = 0$

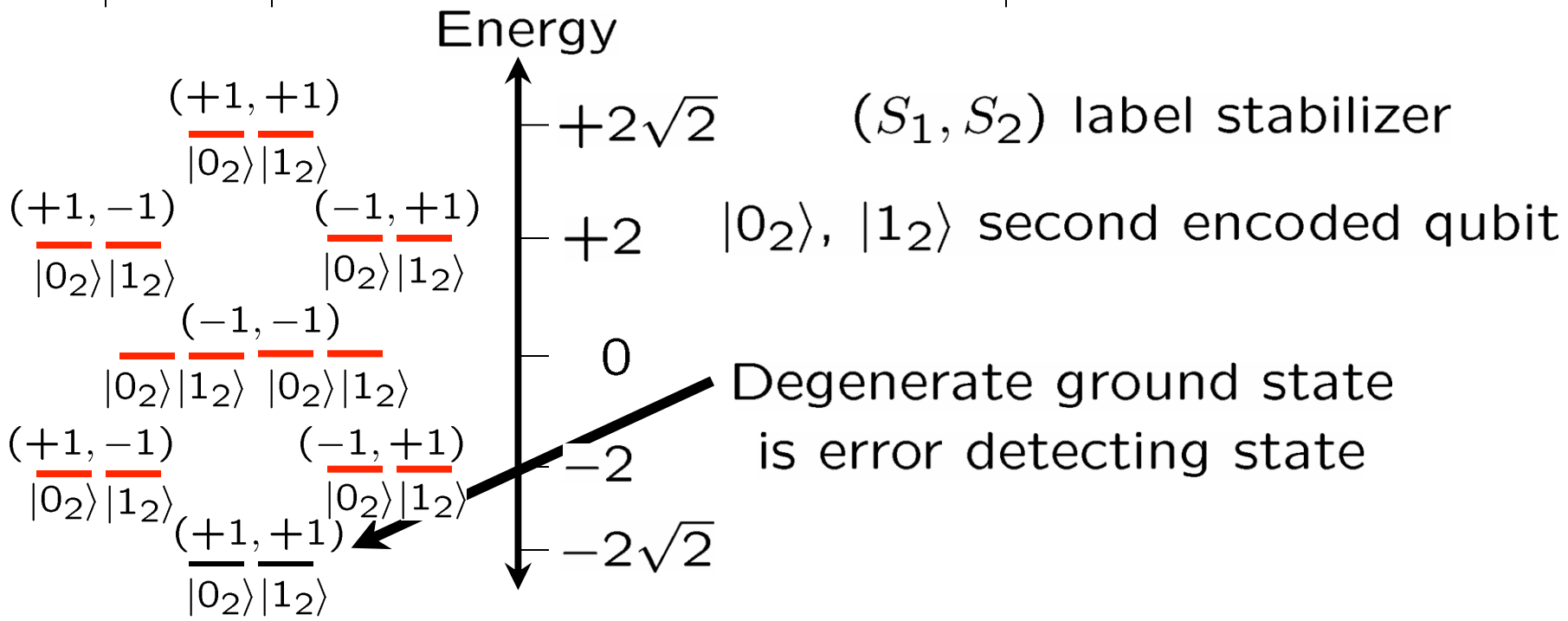
$$H = X_1(I + S_1) + Z_1(I + S_2)$$

Does not involve 2nd encoded qubit.

$\Downarrow$

every energy level at least two-fold degenerate

S_1	S_2	$\mathbf{H} = \mathbf{X}_1(1+S_1) + \mathbf{Z}_1(1+S_2)$	Eigenvalues
$+1$	$+1$	$2(\mathbf{X}_1 + \mathbf{Z}_1)$	$\pm 2\sqrt{2}$
$+1$	-1	$2\mathbf{X}_1$	± 2
-1	$+1$	$2\mathbf{Z}_1$	± 2
-1	-1	$\mathbf{0}$	0



Supercoherence

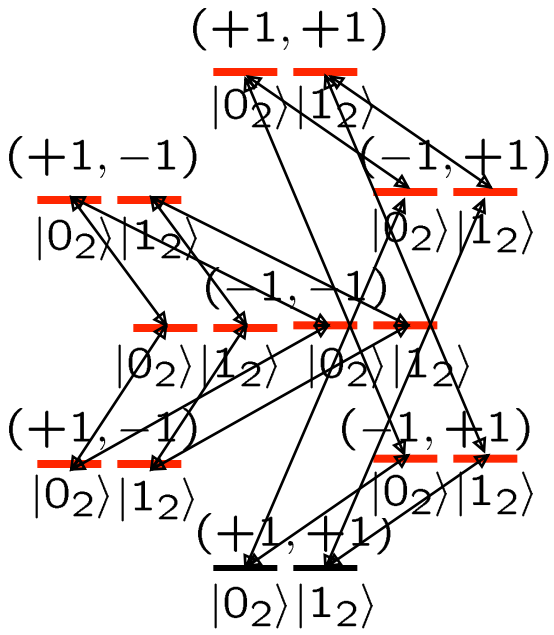
$$\mathbf{H} = \mathbf{X}_1(1 + S_1) + \mathbf{Z}_1(1 + S_2)$$

$\mathbf{S}_i\mathbf{P} = -\mathbf{P}\mathbf{S}_i$ for some i $\quad$ $\mathbf{P}$ single qubit operator

ALL single qubit errors take degenerate ground state to higher energy levels.

Single qubit errors change value of (S1,S2) and hence take ground state to higher energy level.

Effect of **ZIII**

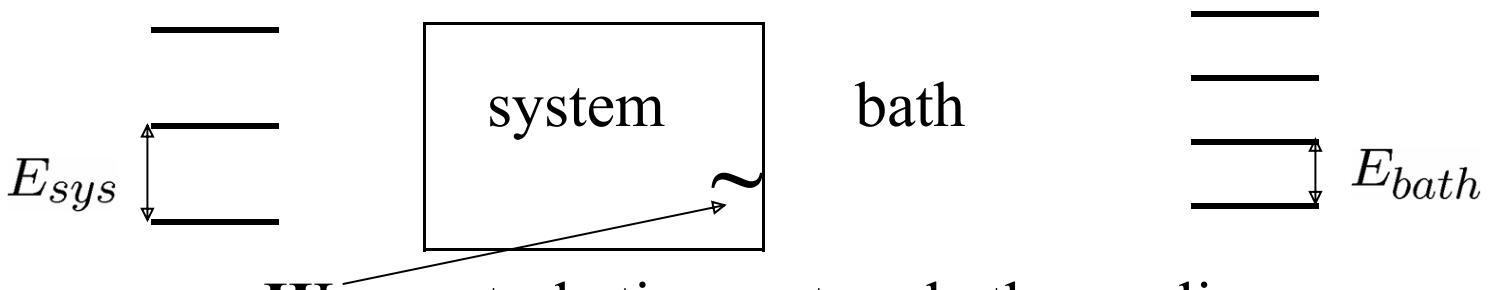


$$\underbrace{(+1, +1)}_{|0_2\rangle \qquad |1_2\rangle}$$

✗ **FORBIDDEN**

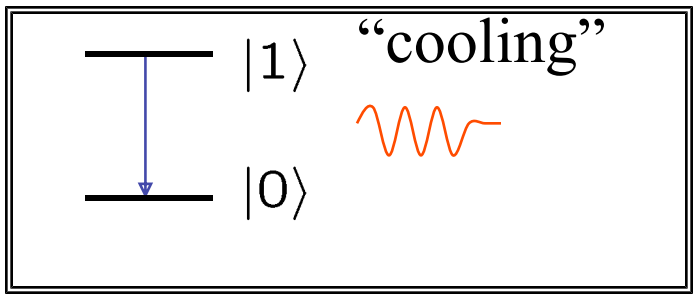
D. Bacon, Ph.D. thesis, U.C. Berkeley, 2001

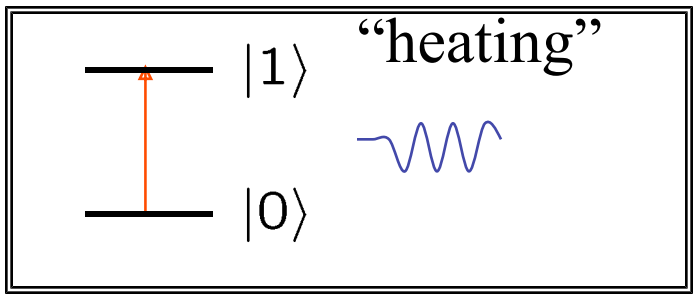
Making Decoherence Work Hard

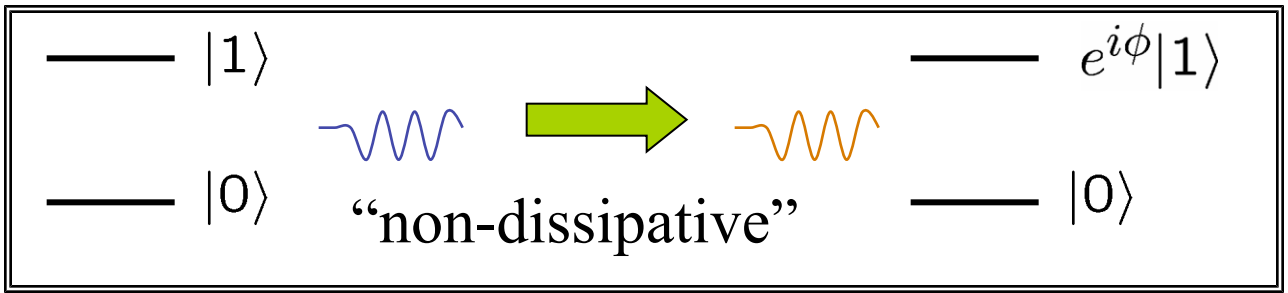


HI = perturbative system-bath coupling

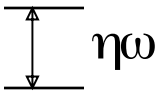
HI energy small compared to Esys and Ebath implies decoherence dominated by pathways that *conserve unperturbed energies* (essentially rotating wave approximation)







two-level atom radiatively coupled to a thermal reservoir:



$$\frac{\partial \rho}{\partial t} = -\bar{n}_{th}\frac{\gamma}{2}\left[\sigma_-\sigma_+\rho + \rho\sigma_-\sigma_+ - 2\sigma_+\rho\sigma_-\right]$$

$$-(\bar{n}_{th}+1)\frac{\gamma}{2}\left[\sigma_+\sigma_-\rho + \rho\sigma_+\sigma_- - 2\sigma_-\rho\sigma_+\right]$$

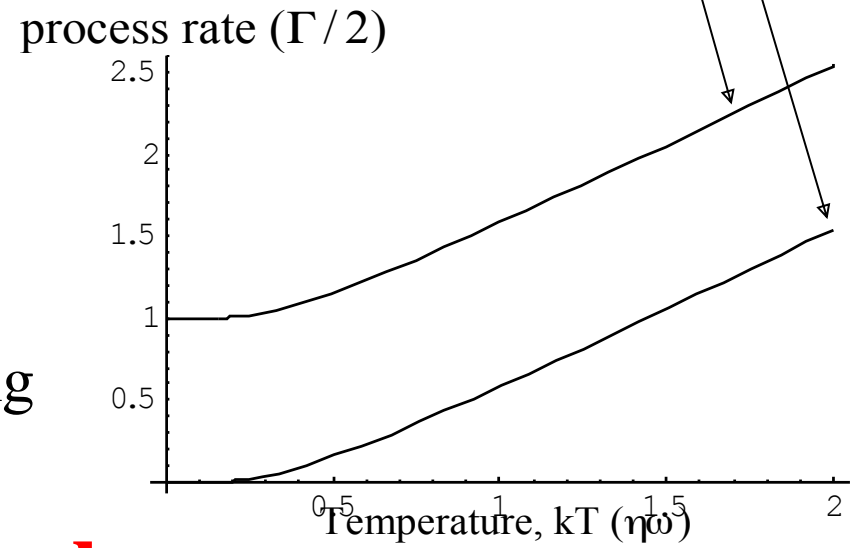
$$-\phi\left[\rho - \sigma_z\rho\sigma_z\right]$$

"heating"

"cooling"

"non-dissipative"

$$\bar{n}_{th} = \frac{1}{\exp\left(\frac{\eta\omega}{kT}\right) - 1}$$



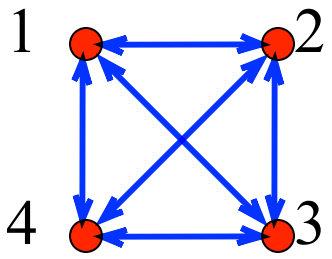
At low bath temperatures, heating disappears……..

Supercoherence made all quantum errors "heating"!

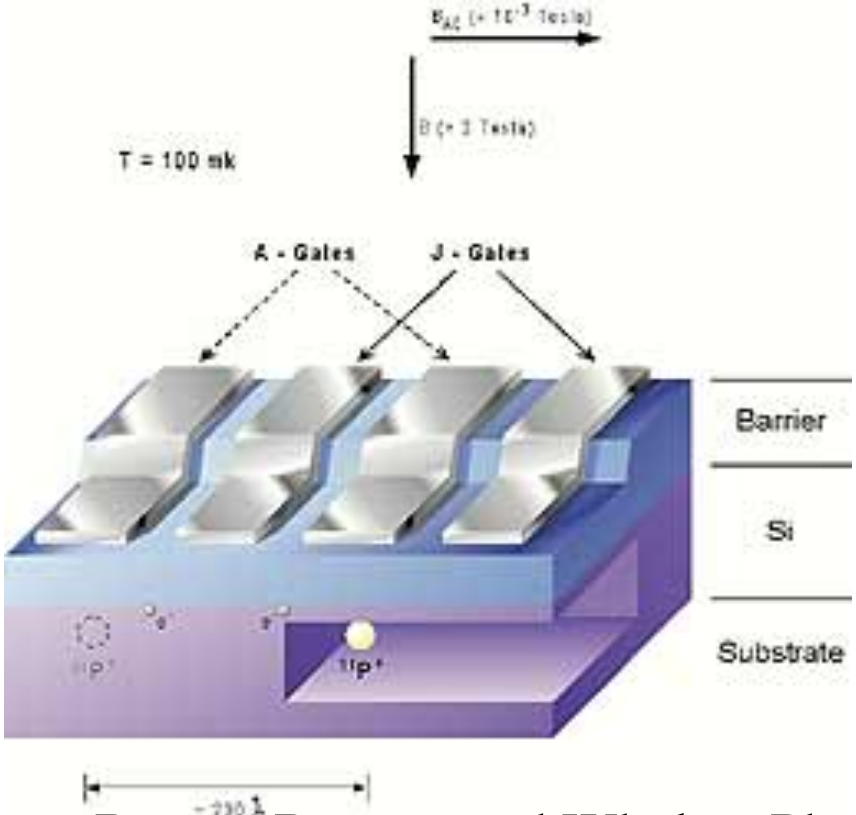
In a Few Years Time…

Four spin systems with equal strength exchange interactions between all spins:

$$\mathbf{H} = \frac{\Delta}{2}\sum_{i,j=1}^{4} \hat{s}_i \cdot \hat{s}_j = \frac{\Delta}{2}\hat{\mathbf{S}}^2 + c\mathbf{I}$$



Supercoherent qubit



single P atoms in Silicon

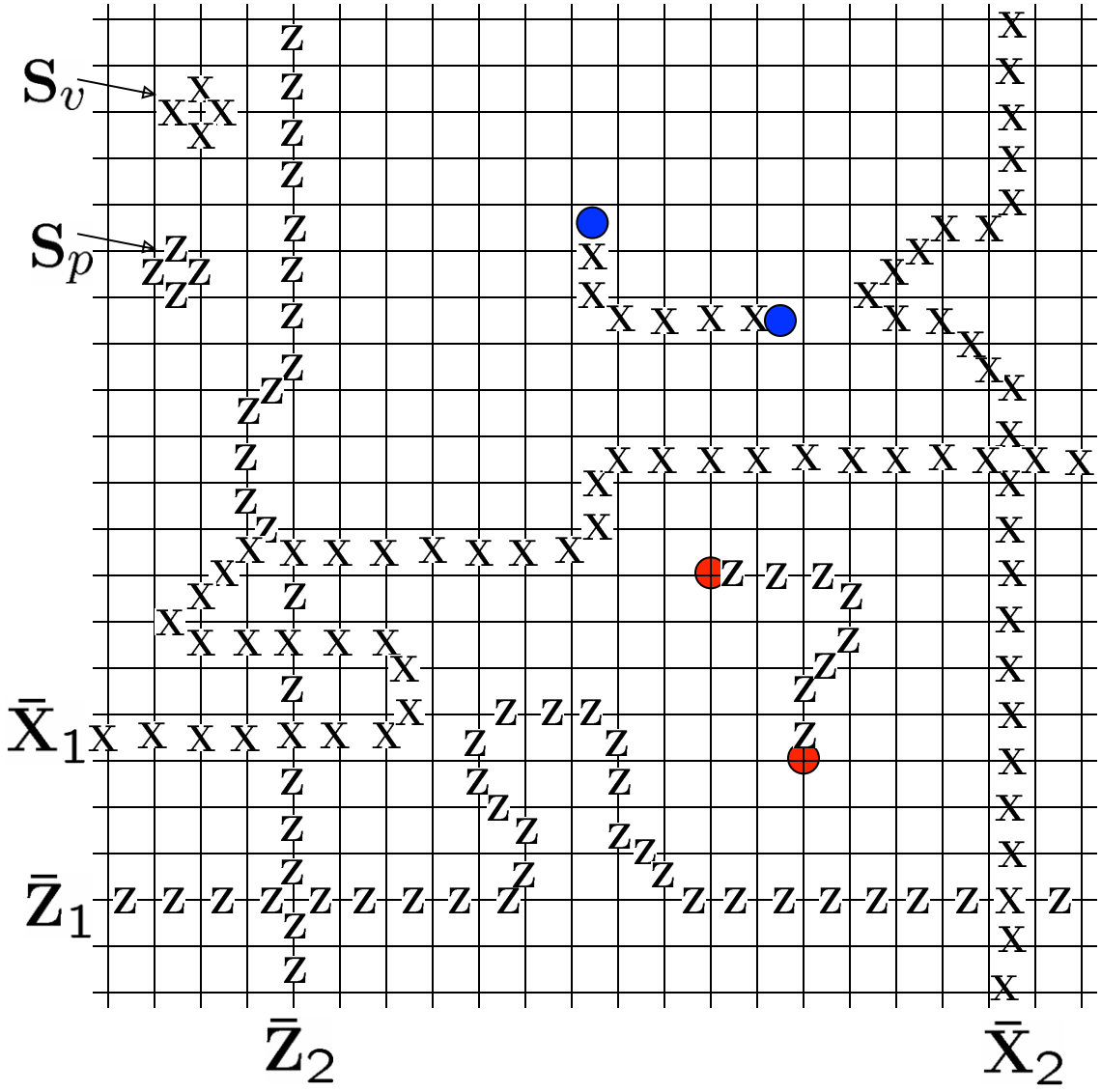
exchange controlled by voltage mediated barriers

Bacon, Brown, and Whaley, Physical Review Letter **87**, 247902 (2001)

DiVincenzo, Bacon, Kempe, Burkard, and Whaley, Nature **408,** 399 (2001)

Supercoherence As the First Step To Achieving the Threshold For Fault-Tolerant Quantum Computation?

Kitaev's Toric Codes



$$\mathbf{S}_p = \sum_{e \in p} \mathbf{Z}_e$$

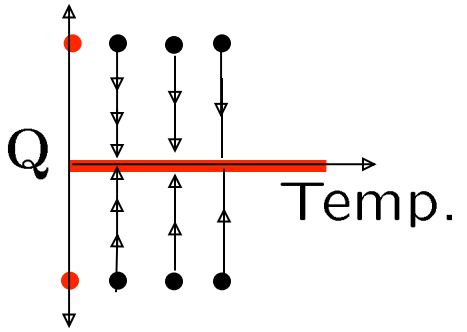
$$\mathbf{S}_v = \sum_{e \in v} \mathbf{X}_e$$

$$\mathbf{H} = -\frac{\Delta}{2}\left(\sum_p \mathbf{S}_p + \sum_v \mathbf{S}_v\right)$$

energy gap $= \Delta$

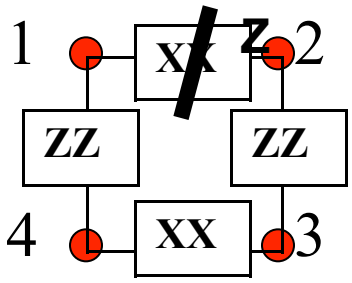
can correct almost any $O(L^2)$ qubit errors (percolation)

natural evolution is not self correcting (we have to work in 2D)



4D Toric code should be self correcting!

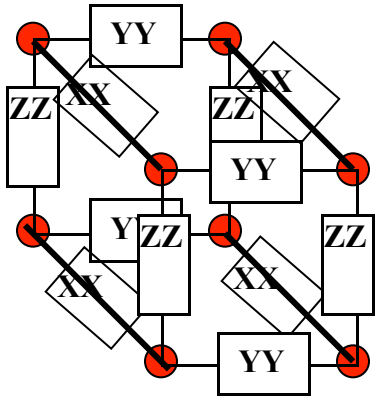
Towards Natural Fault-Tolerance



Error of one type violates bond

$n \times n \times n$

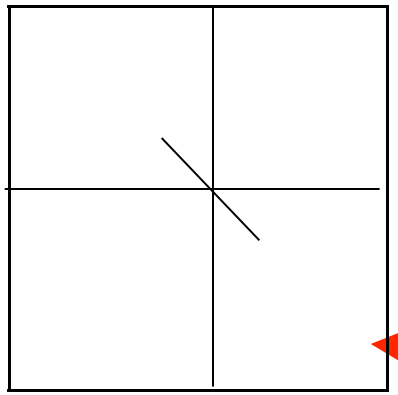
n odd



Cubic lattice with qubits at vertices

$$\mathbf{H} = -J \sum_{ijk} (\mathbf{X}_{i,j,k}\mathbf{X}_{i+1,j,k} + \mathbf{Y}_{i,j,k}\mathbf{Y}_{i,j+1,k} + \mathbf{Z}_{i,j,k}\mathbf{Z}_{i,j,k+1})$$

Logical Operators



$$\bar{\mathbf{Z}} = \mathbf{Z}^{\otimes n^2}$$

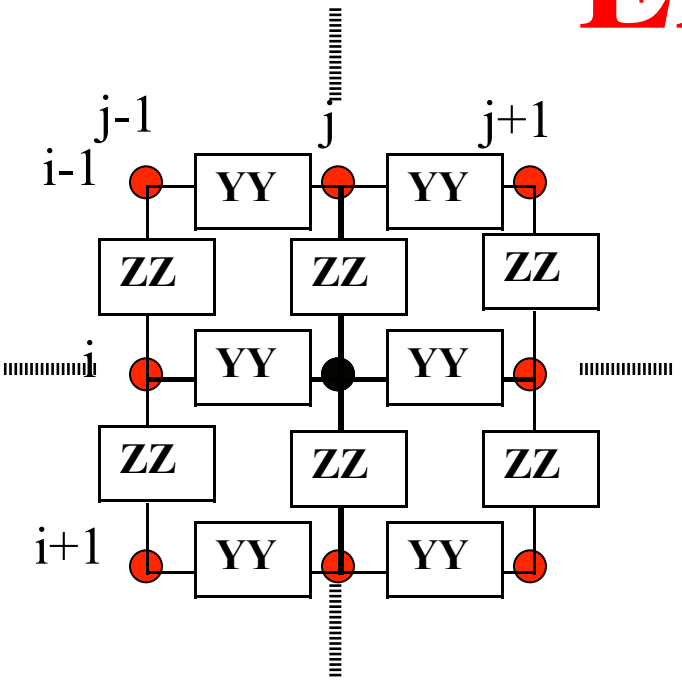
$$\bar{\mathbf{X}} = \mathbf{X}^{\otimes n^2}$$

$$\bar{\mathbf{Y}} = \mathbf{Y}^{\otimes n^2}$$

$$[\bar{\mathbf{X}}, \mathbf{H}] = [\bar{\mathbf{Y}}, \mathbf{H}] = [\bar{\mathbf{Z}}, \mathbf{H}] = 0$$

One encoded qubit

Errors

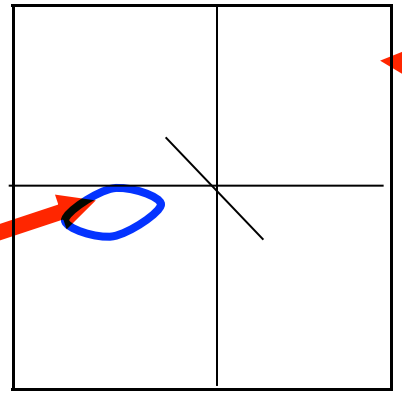


σ_z error

σ_y errors stringlike

σ_x errors stringlike

σ_z errors stringlike



Summary

Learn from our past!

Let physics do the work!

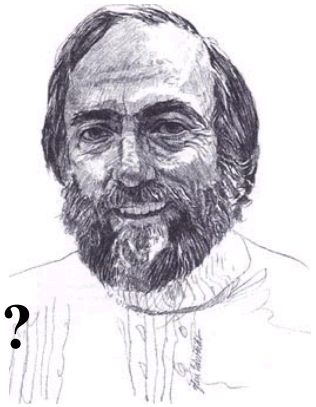
Supercoherence as first step?

Towards self-correcting quantum memories

Is Nature Fault-Tolerant?



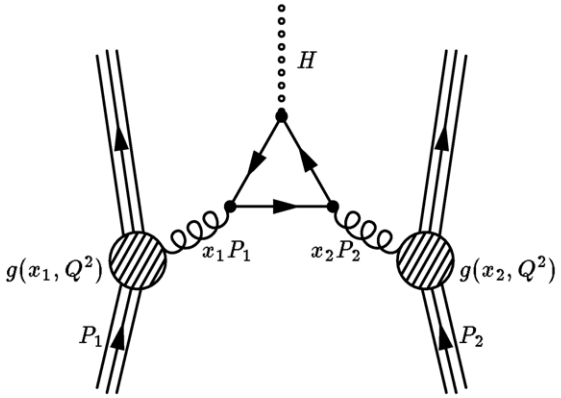
Information paradox for blackholes

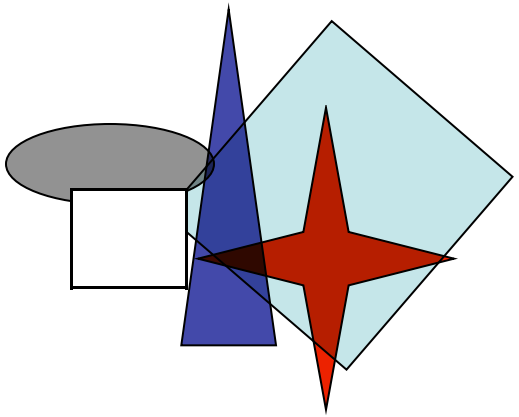


Information is destroyed by a blackhole?

Stephen Hawking

Kip Thorne





John Preskill: "Could the universe be losing information constantly and yet appear to be unitary and quantum information preserving at our scales?"

Fin

Error correcting codes

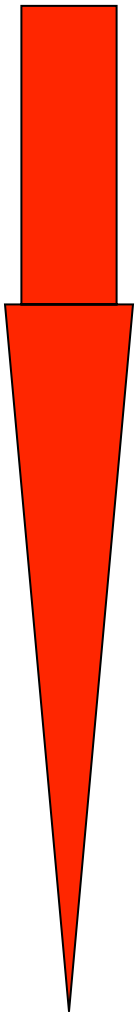
- classical
- quantum

Fault-tolerance

- local structures giving rise to global robustness (CA)

Representation theory

- identification of qubits and logical operations



Geometry, Topology?

Statistical Physics

- theory of strongly interacting quantum systems
- renormalization group
- methods

Real Physics

- physical implementation
- optical lattices, donor electrons, superconducting circuits, etc.

Naturally Fault-Tolerant Quantum Computation